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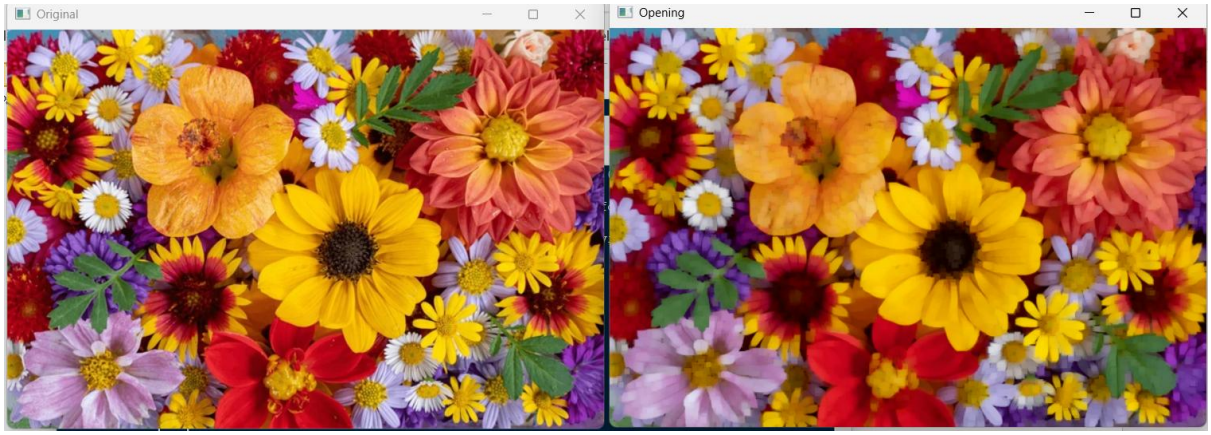
EXPERIMENT 21

CODE:

```
import cv2
import numpy as np
img = cv2.imread("sample.jpg")
kernel = np.ones((5,5), np.uint8)
opening = cv2.morphologyEx(img, cv2.MORPH_OPEN, kernel)
cv2.imshow("Original", img)
cv2.imshow("Opening", opening)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output:



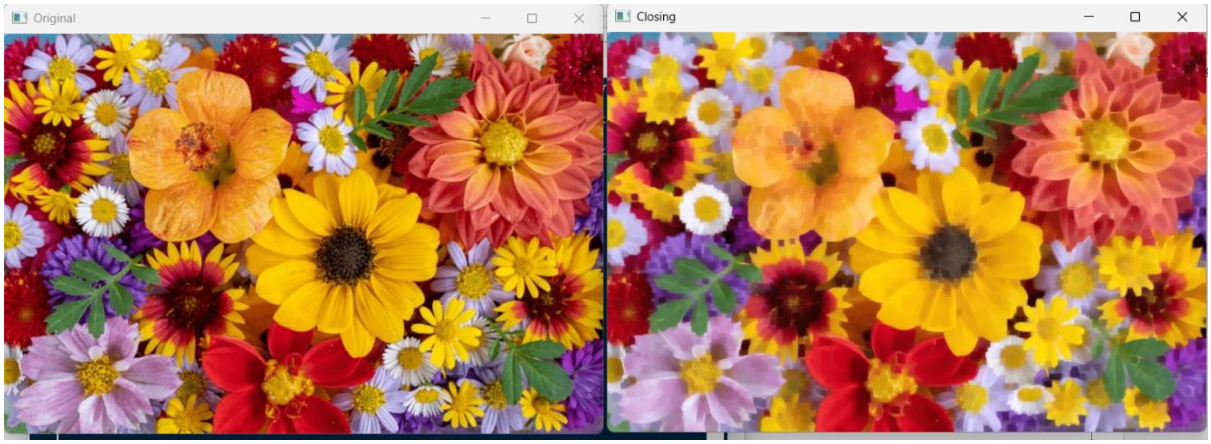
EXPERIMENT 22

CODE:

```
import cv2
import numpy as np
img = cv2.imread("sample.jpg")
```

```
kernel = np.ones((5,5), np.uint8)

closing = cv2.morphologyEx(img, cv2.MORPH_CLOSE, kernel)

cv2.imshow("Original", img)

cv2.imshow("Closing", closing)

cv2.waitKey(0)

cv2.destroyAllWindows()
```

Output:

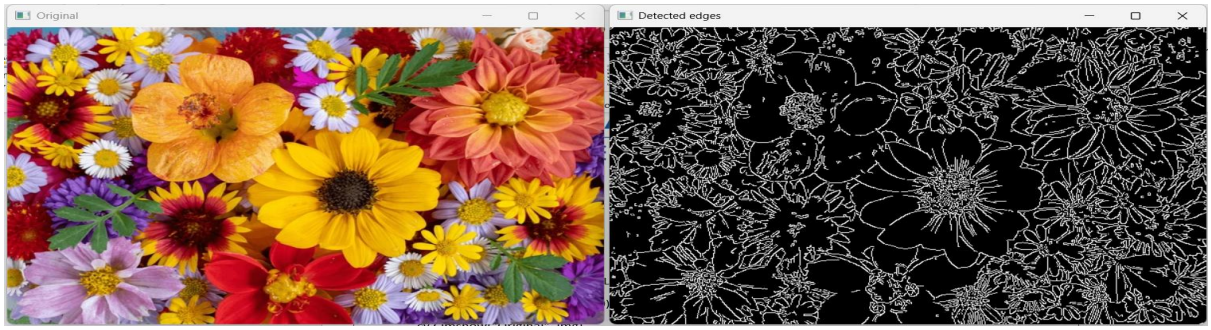


EXPERIMENT 23

CODE:

```
import cv2

img = cv2.imread("sample.jpg")

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

edges = cv2.Canny(gray,100,200)

cv2.imshow("Original", img)

cv2.imshow("Detected edge", edges)

cv2.waitKey(0)

cv2.destroyAllWindows()
```

Output:

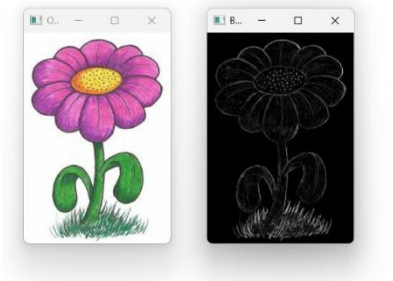


EXPERIMENT 24

Code :

```
import cv2
import numpy as np
img = cv2.imread("sample.jpg")
kernel = np.ones((5,5), np.uint8)
opening = cv2.morphologyEx(img, cv2.MORPH_OPEN, kernel)
cv2.imshow("Original", img)
cv2.imshow("Opening", opening)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

OUTPUT :

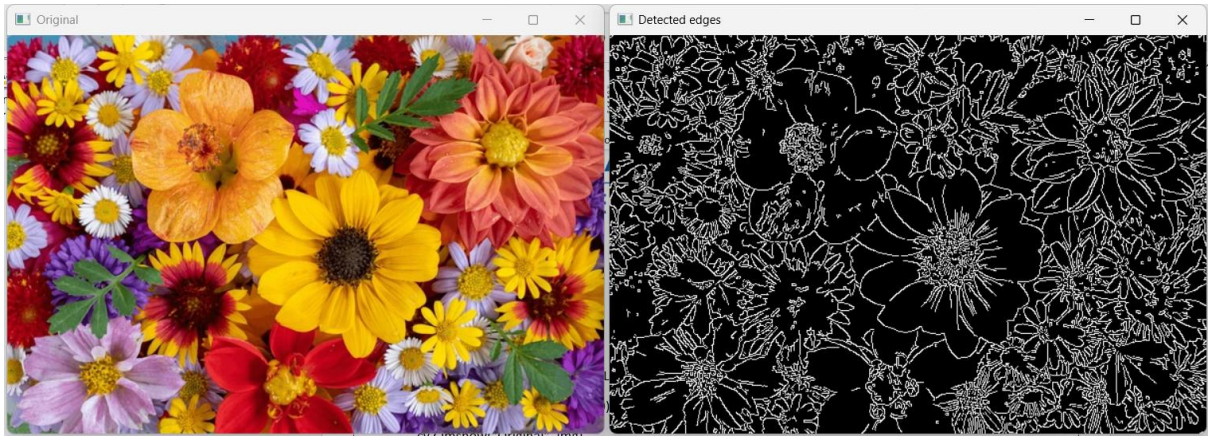


EXPERIMENT 25

CODE:

```
import cv2
img = cv2.imread("sample.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
edges = cv2.Canny(gray,100,200)
cv2.imshow("Original", img)
cv2.imshow("Detected edge", edges)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output:



EXPERIMENT 26

CODE:

```
import cv2

cap = cv2.VideoCapture("video.mp4")

frames = []

while True:

ret, frame = cap.read()

if not ret:

break

frames.append(frame)

for frame in reversed(frames):

cv2.imshow("Reverse Video", frame)

if cv2.waitKey(30) & 0xFF == ord('q'):

break

cap.release()

cv2.destroyAllWindows()
```

Output:

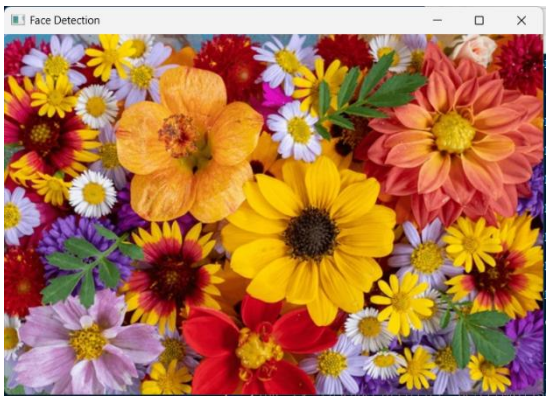
20260310-0753-20.7111837.mp4

EXPERIMENT 27

CODE:

```
import cv2

face = cv2.CascadeClassifier(cv2.data.haarcascades + "haarcascade_frontalface_default.xml")

img = cv2.imread("sample.jpg")

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

faces = face.detectMultiScale(gray, 1.3, 5)

for (x,y,w,h) in faces:

    cv2.rectangle(img,(x,y),(x+w,y+h),(255,0,0),2)


cv2.imshow("Face Detection", img)

cv2.waitKey(0)

cv2.destroyAllWindows()
```

Output:

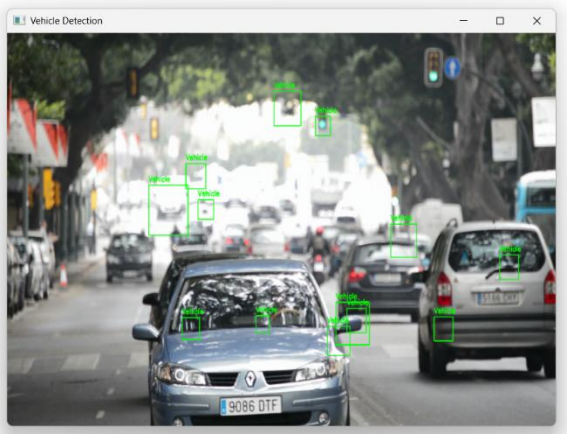


There is no face detected

EXPERIMENT 28

```
import cv2

# Load vehicle cascade classifier
car_cascade = cv2.CascadeClassifier("cars.xml")

# Open input video
cap = cv2.VideoCapture(r"C:\Users\DEEKSHANA PRABHU\Downloads\vehicle.mov")

if not cap.isOpened():
    print("Error opening video")
    exit()

while True:
    ret, frame = cap.read()

    if not ret:
        break

    # Convert frame to grayscale
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    # Detect vehicles
    cars = car_cascade.detectMultiScale(gray, 1.1, 3)

    # Draw rectangles around vehicles
    for (x, y, w, h) in cars:
        cv2.rectangle(frame, (x, y), (x+w, y+h), (0,255,0), 2)
        cv2.putText(frame, "Vehicle", (x, y-10),
                    cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0,255,0), 2)

    # Resize output window
    frame = cv2.resize(frame, (700, 500))

    # Show output
    cv2.imshow("Vehicle Detection", frame)

    if cv2.waitKey(30) & 0xFF == 27:
        break

cap.release()
cv2.destroyAllWindows()
```

OUTPUT :



EXPERIMENT 29

CODE:

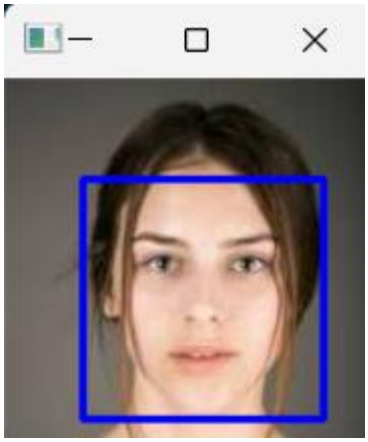
```
import cv2
car_cascade = cv2.CascadeClassifier(
    "C:/Users/RACHITHA/Desktop/computer vision/cars.xml"
)
if car_cascade.empty():
    print("Error: cars.xml file not loaded. Check file path.")
    exit()
cap = cv2.VideoCapture("C:/Users/RACHITHA/Desktop/computer vision/video1.mp4")
if not cap.isOpened():
    print("Error: Video not found")
    exit()
while True:
    ret, frame = cap.read()
    if not ret:
        break
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
    cars = car_cascade.detectMultiScale(gray, 1.1, 3)
    for (x, y, w, h) in cars:
        cv2.rectangle(frame, (x, y), (x+w, y+h), (0, 255, 0), 2)
       cv2.imshow("Vehicle Detection", frame)
    if cv2.waitKey(30) & 0xFF == 27:
        break

cap.release()
cv2.destroyAllWindows()
```

Output:

20260310-0921-36.2046379.mp4

EXPERIMENT 30

CODE:

```
import cv2
face = cv2.CascadeClassifier(cv2.data.haarcascades + "haarcascade_frontalface_default.xml")
smile = cv2.CascadeClassifier(cv2.data.haarcascades + "haarcascade_smile.xml")
img = cv2.imread("sample.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
faces = face.detectMultiScale(gray,1.3,5)
for (x,y,w,h) in faces:
    cv2.rectangle(img,(x,y),(x+w,y+h),(255,0,0),2)
    roi_gray = gray[y:y+h, x:x+w]
    roi_color = img[y:y+h, x:x+w]
    smiles = smile.detectMultiScale(roi_gray,1.8,20)
    for (sx,sy,sw,sh) in smiles:
        cv2.rectangle(roi_color,(sx,sy),(sx+sw,sy+sh),(0,255,0),2)
cv2.imshow("Smile Detection", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output :